In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
iris = sns.load_dataset('iris')
print(iris.head())
print("\nShape:", iris.shape)
print("\nColumns:", iris.columns.tolist())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa

Shape: (150, 5)

Columns: ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'specie
s']
```

In [3]:
```
print("Features and their types:")
print(iris.dtypes)
print("\nUnique species (nominal):", iris['species'].unique())
```

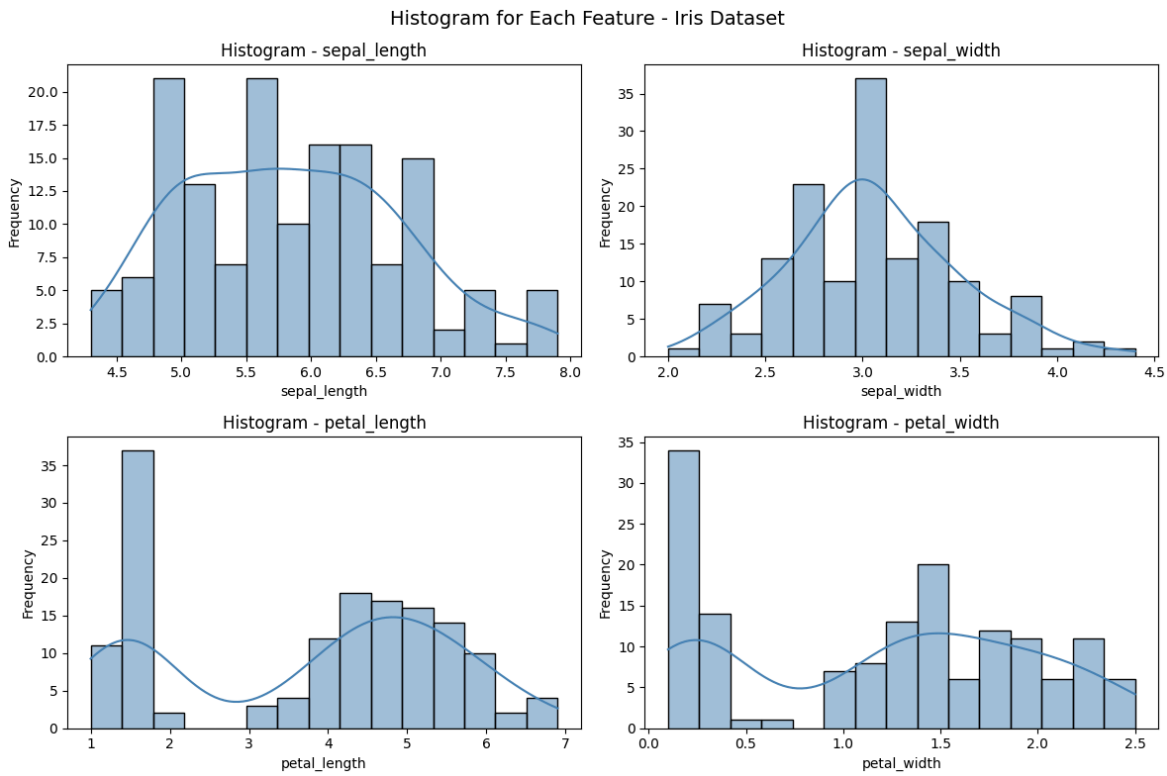
```
Features and their types:
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species             str
dtype: object

Unique species (nominal): <StringArray>
['setosa', 'versicolor', 'virginica']
Length: 3, dtype: str
```

In [4]:
```
features = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width']

plt.figure(figsize=(12, 8))
for i, feature in enumerate(features):
    plt.subplot(2, 2, i+1)
    sns.histplot(iris[feature], kde=True, bins=15, color='steelblue')
    plt.title(f'Histogram - {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')

plt.suptitle('Histogram for Each Feature - Iris Dataset', fontsize=14)
plt.tight_layout()
plt.show()
```



In [5]:

```
plt.figure(figsize=(12, 8))
for i, feature in enumerate(features):
    plt.subplot(2, 2, i+1)
    ax = sns.boxplot(x='species', y=feature, data=iris,
                palette={'setosa': 'steelblue',
                         'versicolor': 'tomato',
                         'virginica': 'orange'})
    plt.title(f'Box Plot - {feature}')
    plt.xlabel('Species')
    plt.ylabel(feature)

plt.suptitle('Box Plot for Each Feature - Iris Dataset', fontsize=14)
plt.tight_layout()
plt.show()
```

```
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\3188180801.py:4: Fut
ureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  ax = sns.boxplot(x='species', y=feature, data=iris,
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\3188180801.py:4: Fut
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ct.

  ax = sns.boxplot(x='species', y=feature, data=iris,
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\3188180801.py:4: Fut
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C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\3188180801.py:4: Fut
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ct.

  ax = sns.boxplot(x='species', y=feature, data=iris,
```
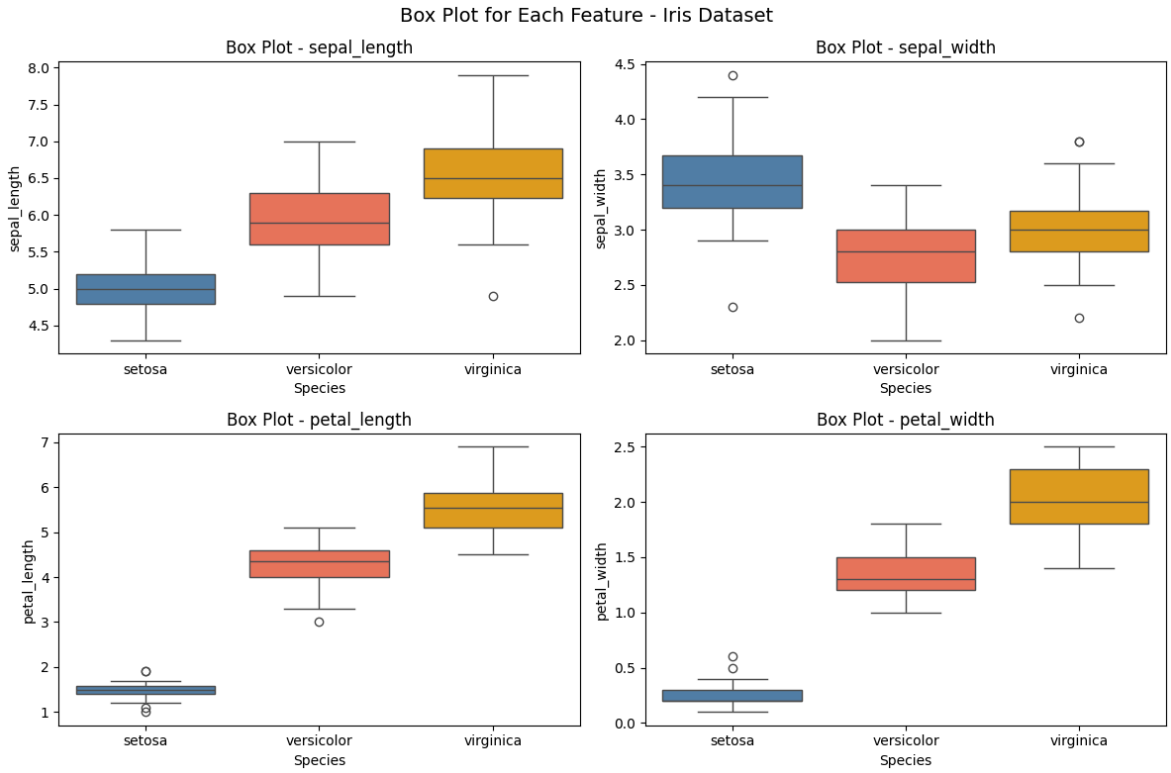
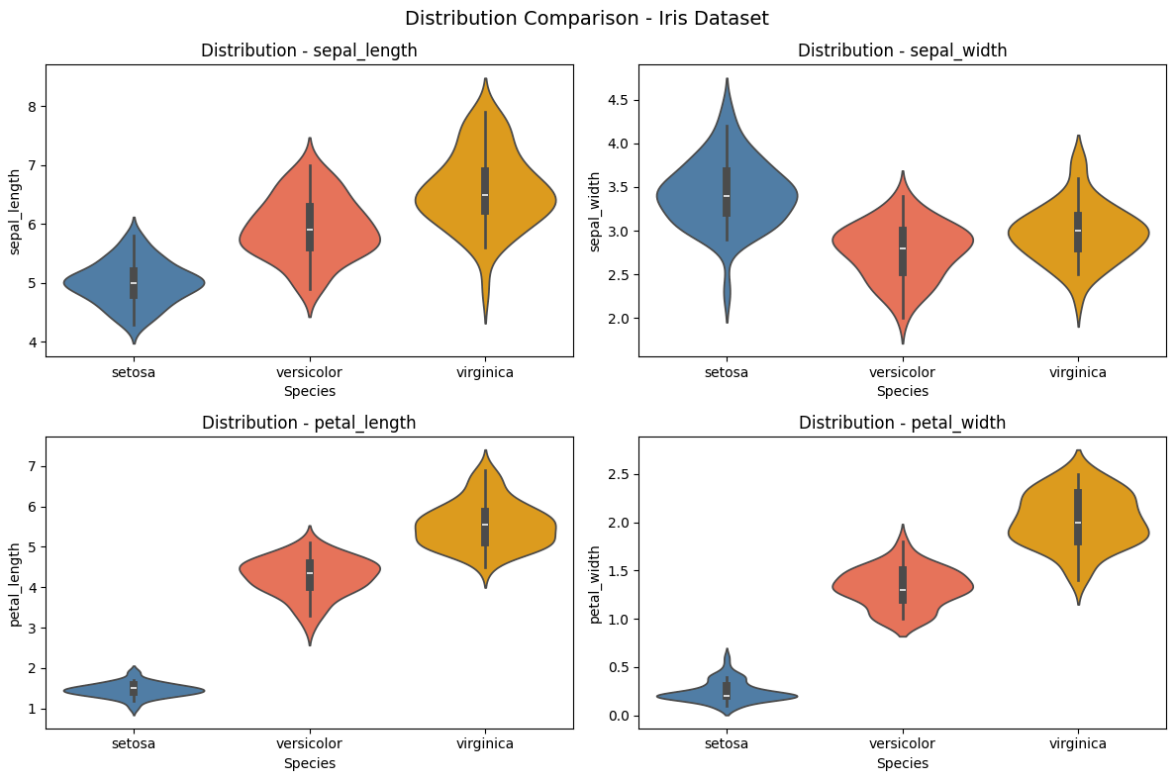



In [6]:
```
plt.figure(figsize=(12, 8))
for i, feature in enumerate(features):
    plt.subplot(2, 2, i+1)
```

```
    sns.violinplot(x='species', y=feature, data=iris,
                   palette={'setosa': 'steelblue',
                            'versicolor': 'tomato',
                            'virginica': 'orange'})
    plt.title(f'Distribution - {feature}')
    plt.xlabel('Species')
    plt.ylabel(feature)

plt.suptitle('Distribution Comparison - Iris Dataset', fontsize=14)
plt.tight_layout()
plt.show()
```

```
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\2515335008.py:4: Fut
ureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.violinplot(x='species', y=feature, data=iris,
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\2515335008.py:4: Fut
ureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.violinplot(x='species', y=feature, data=iris,
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\2515335008.py:4: Fut
ureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.violinplot(x='species', y=feature, data=iris,
C:\Users\Jaydeep Kulkarni\AppData\Local\Temp\ipykernel_52736\2515335008.py:4: Fut
ureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.violinplot(x='species', y=feature, data=iris,
```



In [7]:
```
print("Outlier Summary")
for feature in features:
    Q1 = iris[feature].quantile(0.25)
    Q3 = iris[feature].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = iris[(iris[feature] < lower) | (iris[feature] > upper)]
    print(f"\n{feature}:")
    print(f"  Lower bound: {lower:.2f}, Upper bound: {upper:.2f}")
    print(f"  Number of outliers: {len(outliers)}")
    if len(outliers) > 0:
        print(f"  Outlier values: {outliers[feature].values}")
```

```
Outlier Summary

sepal_length:
  Lower bound: 3.15, Upper bound: 8.35
  Number of outliers: 0

sepal_width:
  Lower bound: 2.05, Upper bound: 4.05
  Number of outliers: 4
  Outlier values: [4.4 4.1 4.2 2. ]

petal_length:
  Lower bound: -3.65, Upper bound: 10.35
  Number of outliers: 0

petal_width:
  Lower bound: -1.95, Upper bound: 4.05
  Number of outliers: 0
```

Observation:

From the histograms and box plots of the Iris dataset, it is observed that the dataset has four numeric features namely sepal length, sepal width, petal length, and petal width, and one categorical feature species having three types Setosa, Versicolor, and Virginica. From the box plots, Setosa clearly has much smaller petal length and petal width compared to Versicolor and Virginica. Sepal width has the most number of outliers which can be seen as dots outside the whiskers of the box plot. Petal length and petal width show very less overlap between species, making them the most useful features to distinguish between the three species. Overall, petal features are more helpful in identifying the species compared to sepal features.